

Aileron Direct-Drive Pin Load Calculation - Revised Safety Factor

Updated preliminary calculation using SF = 1.2 and SF = 1.5 maximum

1. Purpose and update

This document updates the previous aileron direct-drive pin calculation by reducing the mechanical safety factor. The aerodynamic loads are unchanged. The only changed part is the conversion from hinge moment to design torque and the resulting pin-sizing checks.

The calculation is still preliminary because the exact hinge-moment coefficient for the specific NACA 2412 aileron geometry is not available. Therefore the force-arm method is used for first sizing. Final validation should use control-surface data, CFD, wind-tunnel data, or a physical load test.

For the direct-drive pin concept, the pin torque is approximately the aileron hinge moment. There is no servo arm / aileron horn ratio. The design torque is therefore calculated as $T_{\text{design}} = SF * HM_a / \eta$.

Table 1. Input values and assumptions.

Property	Symbol	Value	Unit	Note
Air density	ρ	1.17	kg/m ³	From current design table
Sizing speed	V	24	m/s	Use Vmax if the aircraft can fly faster
Stall speed	V _{stall}	14	m/s	Comparison only
Mean wing chord	c	0.300	m	Used for Reynolds number
Kinematic viscosity	ν	1.44e-05	m ² /s	From current design table
Single aileron span	b _a	0.7568	m	From current design table
Single aileron chord	c _a	0.075	m	25% of 0.30 m wing chord
Aileron deflection cases	δ_a	10, 15, 20	deg	20 deg is worst case here
Aerodynamic effectiveness factor	τ	0.5 to 1.0	-	1.0 is conservative
Center-of-pressure arm	x _{cp}	0.4 c _a = 0.030	m	Preliminary assumption
Safety factor range	SF	1.2 and 1.5	-	Revised from previous SF=3
Drive efficiency	η	0.90	-	Direct pin/coupling efficiency estimate
Pin length	L _{pin}	30	mm	User-proposed total pin length

2. Aerodynamic quantities

$$q = 0.5 \rho V^2 = 0.5 \times 1.17 \times 24^2 = 336.96 \text{ Pa}$$

$$q_{\text{stall}} = 0.5 \rho V_{\text{stall}}^2 = 114.66 \text{ Pa}$$

$$q / q_{\text{stall}} = 2.94$$

$$Re = V c / \nu = 24 \times 0.30 / 1.44\text{e-}05 = 500000$$

$$A_a = b_a c_a = 0.7568 \times 0.075 = 0.05676 \text{ m}^2$$

If a different maximum speed is later used, scale all aerodynamic forces, hinge moments, direct-drive torques, and pin loads by $(V_{\max}/24)^2$.

3. Aileron normal force

$$C_{N,a} = 2 \pi \tau \delta_a \quad \text{with } \delta_a \text{ in radians}$$

$$F_{N,a} = q A_a C_{N,a}$$

Table 2. Estimated aerodynamic normal force on one aileron at 24 m/s.

Deflection	$C_{N,a}$ lower	$C_{N,a}$ conservative	Force lower	Force conservative
10 deg	0.548	1.097	10.49 N	20.97 N
15 deg	0.822	1.645	15.73 N	31.46 N
20 deg	1.097	2.193	20.97 N	41.95 N

4. Hinge moment and revised direct-drive torque

$$HM_a = c_h q A_a c_a$$

$$q A_a c_a = 336.96 \times 0.05676 \times 0.075 = 1.4344, \text{ so } HM_a = 1.4344 c_h \text{ Nm}$$

$$HM_a = F_{N,a} x_{cp}, \text{ with } x_{cp} = 0.4 c_a = 0.030 \text{ m}$$

$$T_{\text{design}} = SF \times HM_a / \eta$$

Table 3. Hinge moment and direct-drive design torque for one aileron.

Deflection	HM lower	HM conservative	T_{design} , SF=1.2	T_{design} , SF=1.5
10 deg	0.315 Nm	0.629 Nm	0.839 Nm / 8.6 kg cm	1.049 Nm / 10.7 kg cm
15 deg	0.472 Nm	0.944 Nm	1.258 Nm / 12.8 kg cm	1.573 Nm / 16.0 kg cm
20 deg	0.629 Nm	1.258 Nm	1.678 Nm / 17.1 kg cm	2.097 Nm / 21.4 kg cm

5. Rolling moment from both ailerons

$$L_a = 2 F_{N,a} y_{\text{bar}}, \text{ with } y_{\text{bar}} = 0.9866 \text{ m}$$

Table 4. Approximate rolling moment generated by both ailerons at 24 m/s.

Deflection	Rolling moment lower	Rolling moment conservative
10 deg	20.69 Nm	41.39 Nm
15 deg	31.04 Nm	62.08 Nm
20 deg	41.39 Nm	82.77 Nm

6. Revised pin torsion and twist

Worst revised pin torque uses 20 deg conservative case with SF=1.5: $T = 2.097 \text{ Nm}$

$$\tau_{\max} = 16 T / (\pi d^3)$$

$$d_{\min} = [16 T / (\pi \tau_{\text{allow}})]^{1/3}$$

Table 5. Minimum solid pin diameter for the revised SF=1.5 torque.

Allowable shear stress	Required solid pin diameter
50 MPa	5.98 mm
100 MPa	4.74 mm
150 MPa	4.14 mm
200 MPa	3.77 mm

$$J = \pi d^4 / 32$$

$$\theta = T L / (J G), \text{ with } L = 30 \text{ mm}$$

Table 6. Pin torsional stress and twist under the revised SF=1.5 torque.

Pin diameter	Torsional shear stress	Twist, aluminium $G=26 \text{ GPa}$	Twist, steel $G=79 \text{ GPa}$
3 mm	395.6 MPa	17.44 deg	5.74 deg
4 mm	166.9 MPa	5.52 deg	1.82 deg
5 mm	85.5 MPa	2.26 deg	0.74 deg
6 mm	49.5 MPa	1.09 deg	0.36 deg
8 mm	20.9 MPa	0.34 deg	0.11 deg

7. Bearing pressure check with revised factor

With SF=1.2, the radial design force from the conservative 20 deg case is 50.34 N. For a 5 mm pin and 5 mm bearing contact, $p = 2.01 \text{ MPa}$. For a 6 mm pin and 5 mm bearing contact, $p = 1.68 \text{ MPa}$.

With SF=1.5, the radial design force from the conservative 20 deg case is 62.92 N. For a 5 mm pin and 5 mm bearing contact, $p = 2.52 \text{ MPa}$. For a 6 mm pin and 5 mm bearing contact, $p = 2.10 \text{ MPa}$.

8. Revised conclusion

Table 7. Revised design summary.

Result	Value
One-aileron area	0.05676 m ²
Dynamic pressure at 24 m/s	336.96 Pa
Conservative force on one aileron at 20 deg	41.95 N
Conservative hinge moment at 20 deg	1.258 Nm
Direct-drive design torque, SF=1.2 and $\eta=0.90$	1.678 Nm = 17.1 kg cm
Direct-drive design torque, SF=1.5 and $\eta=0.90$	2.097 Nm = 21.4 kg cm
Theoretical pin diameter at $\tau_{\text{allow}}=100 \text{ MPa}$, SF=1.5	4.74 mm
Practical minimum solid metal pin size	5 mm diameter x 30 mm length
Stiffer option if space allows	6 mm diameter x 30 mm length

With the revised safety factor, the practical minimum pin diameter drops from 6 mm to 5 mm for a metal pin with approximately 100 MPa allowable shear stress. A 6 mm pin remains preferable for stiffness, lower twist, and better tolerance to misalignment or imperfect torque locking.

Important caution: reducing the safety factor to 1.2-1.5 leaves much less tolerance for underestimated hinge moment, higher-than-assumed flight speed, servo stall-torque marketing values, backlash, friction, and manufacturing defects. This is more suitable for prototype sizing than for a robust flight-critical design.